NAME

mbimagelist − parses recursive imagelist structures, performing one or more tasks on the image files referenced in the imagelist(s). The default action is to print out out the complete list of image file paths; other possible tasks include also printing out associated timestamps and camera settings, printing out the full imagelist row format, and printing out processing parameter values embedded in the imagelist structure.

VERSION

Version 5.0

SYNOPSIS

mbimagelist
[
--absolutepaths {**-A**}
--copy=*directory* {**-C***directory*}
--copyhere
--end=*yyyy/mm/dd/hh/mm/ss.ssssss*
--files {**-F**}
--help {**-H**}
--imagelist
--input=*file* {**-I***file*}
--left {**-L**}
--parameters {**-P**}
--right {**-R**}
--settings {**-S**}
--single
--start=*yyyy/mm/dd/hh/mm/ss.ssssss*
--verbose {**-V**}
]

DESCRIPTION

MBimagelist is a utility for parsing imagelist files. Imagelist files, or lists of seafloor photographic images, are used by a number of **MB-System** programs dealing with seafloor photography. These lists may contain references to other imagelists, making them recursive. By default, the program **mbimagelist** outputs each image filename encountered as it parses through the input imagelist tree. The output paths can be relative to the current working directory or absolute. Instead of the plain file path listing, the **−−settings** option can be used to also output the image timestamp, gain, and exposure, or the **−−imagelist** option can be used to reproduce the full eight-column imagelist row format (left and right image paths, left and right timestamps, left and right gains, and left and right exposures) for each image or stereo pair. The **−−start** and **−−end** options can be used to restrict the image listings that are output to those falling within a specified time period; by default all image listings encountered are output.

Imagelist files can also contain processing parameters associated with the images, including specifying a file containing the calibration parameters for the camera rig, specifying a file containing the trajectory (navigation and attitude) of the camera rig as images were collected, specifying a file with an image correction lookup table, and several other parameters. All of these parameters correspond to command line options for **mbphotomosaic** that can also be specified. When **−−parameters** is used together with **−−imagelist**, these processing parameter entries are printed in the same form in which they appear in an imagelist file (e.g. "#PARAMETER --calibration-file=20250325_LASSCameraCalibration.yml"); otherwise they are printed as a human-readable summary line.

This program can be used in shellscripts to read imagelists in the same fashion as **MB-System** programs like **mbphotomosaic** and **mbphotocorrect**. This program can also be used to check and debug complex recursive

imagelist structures.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−absolutepaths {−A}
Causes the image file paths in the output to be converted to absolute paths (by prepending the current working directory to any relative path) rather than being output as they appear in the imagelist. Default: paths are output exactly as they appear in the imagelist.

−−copy=*directory* **{−C***directory***}**
Causes each image file listed in the output to also be copied (using the shell **cp** command) into *directory*.

−−copyhere
Equivalent to **−−copy**, except that the destination directory is the current working directory rather than one specified by the user.

−−end=*yyyy/mm/dd/hh/mm/ss.ssssss*
Sets the end of the time period for which image listings are output. Any image (or, for a stereo pair, any record whose left image time is used) with a timestamp later than this time is skipped. The time is specified as year/month/day/hour/minute/second, with the seconds field allowed to include a fractional part. Default: no end time is applied, so image listings are output through the end of the imagelist.

−−files {−F}
Requests that the plain image file paths be listed, one per line. This is the default output mode when none of **−−parameters**, **−−settings**, or **−−imagelist** is specified.

−−help {−H}
This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−imagelist
Requests that image listings be output in the same eight-column form used by imagelist files themselves:
left-image-path right-image-path left-time_d right-time_d left-gain right-gain left-exposure right-exposure
Image paths are output relative to the current working directory (unless **−−absolutepaths** is also given). If an entry has no left or no right image (a single-camera image, or a stereo pair missing one side), "NULL" is printed in place of the missing path and 0.0 is printed for its associated time, gain, and exposure values. This option is an alternative to **−−files** (the default) and **−−settings**; if **−−imagelist** is given it takes precedence over **−−files** and **−−settings** regardless of the order in which they are given on the command line. See also **−−parameters**.

−−input=*file* **{−I***file***}**
Sets the filename of the imagelist to be parsed. Imagelists are recursive, and may reference other imagelists as well as individual image files. Default: *file* = "imagelist.mb-2".

−−left {−L}
Restricts the listing to images from the left camera: stereo pairs (only the left-side path/time/gain/exposure values are listed; the right side is omitted, or written as **NULL** in **−−imagelist** output) and left-only entries. Single-camera (non-stereo) images are excluded. The end-of-run left/right/single/total image counts, and any file copying requested with **−−copy**, are likewise restricted to what is actually listed. Default: every image is listed.

−−parameters {−P}
Causes a line to be printed for each processing parameter entry encountered while parsing the imagelist, in addition to (or instead of, if **−−files** and **−−settings** are not also given) the image file listing. If **−−imagelist** is also in use, each parameter line is printed in the same form found in imagelist files, for example:

#PARAMETER --calibration-file=20250325_LASSCameraCalibration.yml

Otherwise the parameter line is printed as a human-readable summary.

For those parameter options whose value is a file path (including but not limited to **−−platform-file**, **−−calibration-file**, **−−navigation-file**, **−−tide-file**, **−−image-quality-file**, and **−−correction-file**), a relative path value is rewritten to be prefixed by the directory containing the imagelist file in which the **#PARAMETER** entry was found, so that the path output is valid relative to the current working directory. This rewriting is applied regardless of which of **−−files**, **−−settings**, or **−−imagelist** is in effect, and follows the same convention used by **mbphotomosaic** when it reads these entries. A path value that is already absolute (beginning with '/') is left unmodified. Default: parameter entries are not reported.

−−right {−R}
Restricts the listing to images from the right camera: stereo pairs (only the right-side path/time/gain/exposure values are listed; the left side is omitted, or written as **NULL** in **−−imagelist** output) and right-only entries. Single-camera (non-stereo) images are excluded. The end-of-run left/right/single/total image counts, and any file copying requested with **−−copy**, are likewise restricted to what is actually listed. Default: every image is listed.

−−settings {−S}
Causes each image line in the output to include the image timestamp, gain, and exposure values, in addition to the file path. Default: only the file path is printed. This option is an alternative to **−−files** (the default); see **−−imagelist** for a further alternative output form.

−−single
Restricts the listing to single-camera (non-stereo) images, excluding stereo pairs and left-only or right-only entries. The end-of-run left/right/single/total image counts, and any file copying requested with **−−copy**, are likewise restricted to what is actually listed. Default: every image is listed. There is no short option equivalent.

−−start=*yyyy/mm/dd/hh/mm/ss.ssssss*
Sets the start of the time period for which image listings are output. Any image (or, for a stereo pair, any record whose left image time is used) with a timestamp earlier than this time is skipped. The time is specified as year/month/day/hour/minute/second, with the seconds field allowed to include a fractional part. Default: no start time is applied, so image listings are output starting from the beginning of the imagelist.

−−verbose {−V}
Increases the verbosity of **mbimagelist**. Can be used multiple times. If used at all, a summary of the numbers of left, right, single, and total images listed is printed at the end of execution.

EXAMPLES

Suppose we have a directory of images collected by a stereo camera rig, and that a file named imagelist.mb-2 has been constructed referencing the corresponding left and right image files along with their timestamps, gains, and exposures, together with the name of the calibration file used to process them. The tag **#STEREOCAMERA** indicates that unpaired image entries further down in the file should be treated as one-sided stereo pairs rather than single-camera images. The contents of imagelist.mb-2 might be:

```
#STEREOCAMERA
#PARAMETER --calibration-file=20250325_LASSCameraCalibration.yml
PROSILICA_L/0001.tif PROSILICA_R/0001.tif 1700000000.123456 1700000000.223456 10.0 11.0 4000.0 4100.0
PROSILICA_L/0002.tif PROSILICA_R/0002.tif 1700003600.123456 1700003600.223456 12.0 13.0 4200.0 4300.0
PROSILICA_L/0003.tif 1700007200.123456
```

The last line above is a single left-camera image with no right-camera counterpart.

If we run:

```
mbimagelist --input=imagelist.mb-2
```

the default **--files** output lists just the image paths encountered, one per line:

```
PROSILICA_L/0001.tif
PROSILICA_R/0001.tif
PROSILICA_L/0002.tif
PROSILICA_R/0002.tif
PROSILICA_L/0003.tif
```

If we run:

```
mbimagelist --input=imagelist.mb-2 --settings
```

each image path is followed by its timestamp, gain, and exposure:

```
PROSILICA_L/0001.tif 1700000000.123456 10.000000 4000.000000
PROSILICA_R/0001.tif 1700000000.223456 11.000000 4100.000000
PROSILICA_L/0002.tif 1700003600.123456 12.000000 4200.000000
PROSILICA_R/0002.tif 1700003600.223456 13.000000 4300.000000
PROSILICA_L/0003.tif 1700007200.123456 15.000000 4000.000000
```

The single left-camera image is reported with the default gain (15.0) and exposure (4000.0) values, since none were given in the imagelist.

If we run:

```
mbimagelist --input=imagelist.mb-2 --imagelist --parameters
```

the output reproduces the imagelist file's own eight-column row format and also reports the processing parameter entry in the same form used in imagelist files:

```
#PARAMETER --calibration-file=20250325_LASSCameraCalibration.yml
PROSILICA_L/0001.tif PROSILICA_R/0001.tif 1700000000.123456 1700000000.223456 10.000000 11.000000 4000.000000 4100.000000
PROSILICA_L/0002.tif PROSILICA_R/0002.tif 1700003600.123456 1700003600.223456 12.000000 13.000000 4200.000000 4300.000000
PROSILICA_L/0003.tif NULL 1700007200.123456 0.000000 15.000000 0.000000 4000.000000 0.000000
```

Since the third entry has no right-camera image, "NULL" is printed for the missing path, and 0.0 is printed for the associated time, gain, and exposure values.

If instead the imagelist file were located in a subdirectory, for example survey1/imagelist.mb-2, and also contained a navigation file parameter given as an absolute path:

```
#PARAMETER --calibration-file=20250325_LASSCameraCalibration.yml
#PARAMETER --navigation-file=/data/survey1/nav.txt
```

then running any of **--files --parameters**, **--settings --parameters**, or **--imagelist --parameters** against survey1/imagelist.mb-2 outputs:

```
#PARAMETER --calibration-file=survey1/20250325_LASSCameraCalibration.yml
#PARAMETER --navigation-file=/data/survey1/nav.txt
```

The relative calibration file path is rewritten to be prefixed with the imagelist file's own directory, so that it remains valid relative to the current working directory, while the absolute navigation file path is left unmodified. This directory-prefixing is applied to the value of any **#PARAMETER** option that takes a file path, in all three of the **--files**, **--settings**, and **--imagelist** output modes.

The timestamps in the first two entries correspond to 2023/11/14 22:13:20 and 2023/11/14 23:13:20 respectively. If we only want image listings collected between 23:00 and 23:30 on that date, we can run:

```
mbimagelist --input=imagelist.mb-2 --imagelist \
        --start=2023/11/14/23/00/00 --end=2023/11/14/23/30/00
```

which yields only the second stereo pair:

```
PROSILICA_L/0002.tif PROSILICA_R/0002.tif 1700003600.123456 1700003600.223456 12.000000 13.000000 4200.000000 4300.000000
```

SEE ALSO

mbsystem(1)

BUGS

No true bugs here, only distantly related arthropods... Yum. Seriously, it would be better if the copy function preserved the modification times of the copied swath files and ancillary files. Copying of processed files should also be an option.